

MERN stack - Shopping Online

Lab 05

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1 MERN stack

1.1 Functionals

1.1.1 Customer - home

1.1.1.1 Server

- Update 'server/models/ProductDAO.js' file

```
1 ...
2 const ProductDAO = {
3   ...,
4   async selectByID(_id) {
5     const product = await Models.Product.findById(_id).exec();
6     return product;
7   },
8   async selectTopNew(top) {
9     const query = {};
10    const mysort = { cdate: -1 }; // descending
11    const products = await Models.Product.find(query).sort(mysort).limit(top).exec();
12    return products;
13  },
14  async selectTopHot(top) {
15    const items = await Models.Order.aggregate([
16      { $match: { status: 'APPROVED' } },
17      { $unwind: '$items' },
18      { $group: { _id: '$items.product._id', sum: { $sum: '$items.quantity' } } },
19      { $sort: { sum: -1 } }, // descending
20      { $limit: top }
21    ]).exec();
22    var products = [];
23    for (const item of items) {
24      const product = await ProductDAO.selectByID(item._id);
25      products.push(product);
26    }
27    return products;
28  }
29 };
30 ...
```

- Create 'server/api/customer.js' file

```
1 const express = require('express');
2 const router = express.Router();
3 // daos
4 const CategoryDAO = require('../models/CategoryDAO');
5 const ProductDAO = require('../models/ProductDAO');
6 // category
7 router.get('/categories', async function (req, res) {
8   const categories = await CategoryDAO.selectAll();
9   res.json(categories);
10 });
11 // product
12 router.get('/products/new', async function (req, res) {
13   const products = await ProductDAO.selectTopNew(3);
14   res.json(products);
15 });
16 router.get('/products/hot', async function (req, res) {
17   const products = await ProductDAO.selectTopHot(3);
18   res.json(products);
19 });
20 module.exports = router;
```

- Update 'server/index.js' file

```
1 ...
2 // apis
3 app.use('/api/customer', require('../api/customer.js'));
```

- Postman:

- (GET) <http://localhost:3000/api/customer/categories>
- (GET) <http://localhost:3000/api/customer/products/new>
- (GET) <http://localhost:3000/api/customer/products/hot>

1.1.1.2 Client-customer

- Create 'client-customer/src/components/MenuComponent.js' file

```
1 import axios from 'axios';
2 import React, { Component } from 'react';
3
4 class Menu extends Component {
5   constructor(props) {
6     super(props);
7     this.state = {
8       categories: []
9     };
10  }
11  render() {
12    const cates = this.state.categories.map((item) => {
13      return (
14        <li key={item._id} className="menu"><a href="">{item.name}</a></li>
15      );
16    });
17    return (
18      <div className="border-bottom">
19        <div className="float-left">
20          <ul className="menu">
21            <li className="menu"><a href="">Home</a></li>
22            {cates}
23          </ul>
24        </div>
25        <div className="float-right">
26          <form className="search">
27            <input type="search" placeholder="Enter keyword" className="keyword" />
28            <input type="submit" value="SEARCH" />
29          </form>
30        </div>
31        <div className="float-clear" />
32      </div>
33    );
34  }
35  componentDidMount() {
36    this.apiGetCategories();
37  }
38  // apis
39  apiGetCategories() {
40    axios.get('/api/customer/categories').then((res) => {
41      const result = res.data;
42      this.setState({ categories: result });
43    });
44  }
45 }
46 export default Menu;
```

- Create 'client-customer/src/components/InformComponent.js' file

```
1 import React, { Component } from 'react';
2
3 class Inform extends Component {
4   render() {
5     return (
6       <div className="border-bottom">
7         <div className="float-left">
8           <a href="">Login</a> | <a href="">Sign-up</a> | <a href="">Active</a>
9         </div>
10      </div>
11    );
12  }
13 }
```

```

9         </div>
10        <div className="float-right">
11            <a href="#">My cart</a> have <b>0</b> items
12        </div>
13        <div className="float-clear" />
14    </div>
15    );
16    }
17 }
18 export default Inform;

```

- Create 'client-customer/src/components/HomeComponent.js' file

```

1  import axios from 'axios';
2  import React, { Component } from 'react';
3
4  class Home extends Component {
5      constructor(props) {
6          super(props);
7          this.state = {
8              newprods: [],
9              hotprods: []
10         };
11     }
12     render() {
13         const newprods = this.state.newprods.map((item) => {
14             return (
15                 <div key={item._id} className="inline">
16                     <figure>
17                         <a href="#"><img src={"data:image/jpg;base64," + item.image} width="300px" height="
18                             300px" alt="" /></a>
19                         <figcaption className="text-center">{item.name}<br />Price: {item.price}</
20                             figcaption>
21                     </figure>
22                 </div>
23             );
24         });
25         const hotprods = this.state.hotprods.map((item) => {
26             return (
27                 <div key={item._id} className="inline">
28                     <figure>
29                         <a href="#"><img src={"data:image/jpg;base64," + item.image} width="300px" height="
30                             300px" alt="" /></a>
31                         <figcaption className="text-center">{item.name}<br />Price: {item.price}</
32                             figcaption>
33                     </figure>
34                 </div>
35             );
36         });
37         return (
38             <div>
39                 <div className="align-center">
40                     <h2 className="text-center">NEW PRODUCTS</h2>
41                     {newprods}
42                 </div>
43                 {this.state.hotprods.length > 0 ?
44                     <div className="align-center">
45                         <h2 className="text-center">HOT PRODUCTS</h2>
46                         {hotprods}
47                     </div>
48                     : <div />}
49             </div>
50         );
51     }
52     componentDidMount() {
53         this.apiGetNewProducts();
54         this.apiGetHotProducts();
55     }
56     // apis

```

```

53   apiGetNewProducts() {
54     axios.get('/api/customer/products/new').then((res) => {
55       const result = res.data;
56       this.setState({ newprods: result });
57     });
58   }
59   apiGetHotProducts() {
60     axios.get('/api/customer/products/hot').then((res) => {
61       const result = res.data;
62       this.setState({ hotprods: result });
63     });
64   }
65 }
66 export default Home;

```

- Create 'client-customer/src/components/MainComponent.js' file

```

1  import React, { Component } from 'react';
2  import Menu from './MenuComponent';
3  import Inform from './InformComponent';
4  import Home from './HomeComponent';
5
6  class Main extends Component {
7    render() {
8      return (
9        <div className="body-customer">
10          <Menu />
11          <Inform />
12          <Home />
13        </div>
14      );
15    }
16  }
17  export default Main;

```

- Update 'client-customer/src/App.js' file

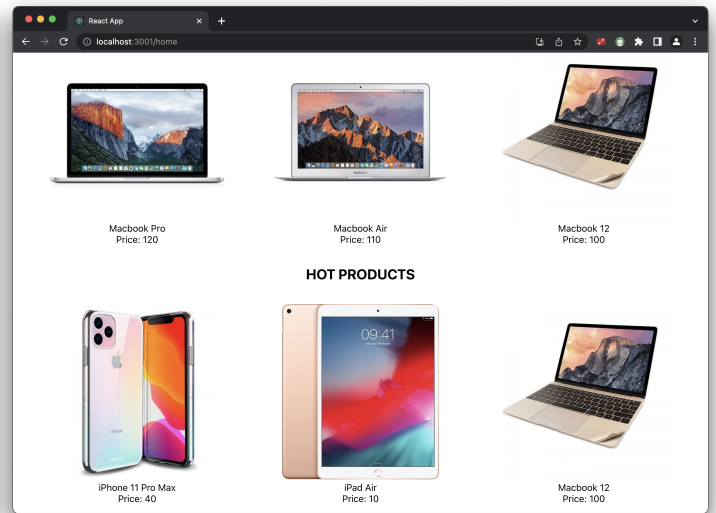
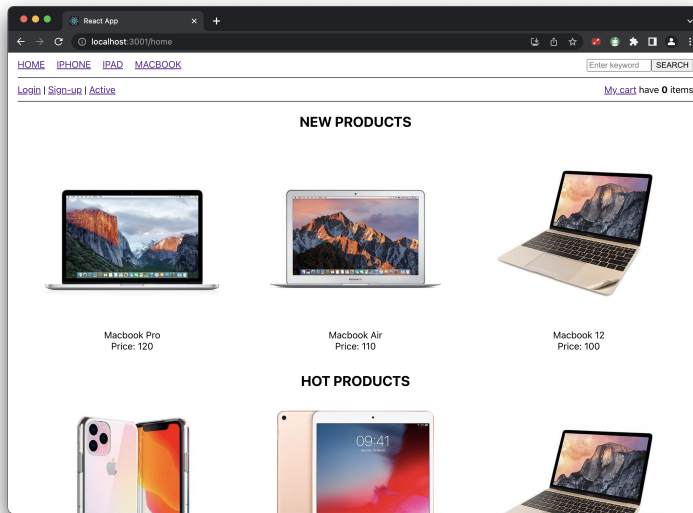
```

1  import './App.css';
2  import React, { Component } from 'react';
3  import Main from './components/MainComponent';
4
5  class App extends Component {
6    render() {
7      return (
8        <Main />
9      );
10   }
11 }
12 export default App;

```

- Browser: <http://localhost:3001/>

1.1.1.3 Screenshots



1.1.1.4 React Router

- Install react-router-dom library.

```
1 $ npm install react-router-dom --save
```

- Update 'client-customer/src/App.js' file

```
1 ...
2 import { BrowserRouter } from 'react-router-dom';
3
4 class App extends Component {
5   render() {
6     return (
7       <BrowserRouter>
8         <Main />
9       </BrowserRouter>
10    );
11  }
12 }
13 ...
```

- Update 'client-customer/src/components/MainComponent.js' file

```
1 ...
2 import { Routes, Route, Navigate } from 'react-router-dom';
3
4 class Main extends Component {
5   ...
6   render() {
7     ...
8     return (
9       <div className="body-customer">
10         <Menu />
11         <Inform />
12         <Routes>
13           <Route path="/" element={<Navigate replace to="/home" />} />
14           <Route path="/home" element={<Home />} />
15         </Routes>
16       </div>
17     );
18     ...
19   }
20 }
21 ...
```

- Update 'client-customer/src/components/MenuComponent.js' file

```

1  ...
2  import { Link } from 'react-router-dom';
3
4  class Menu extends Component {
5    ...
6    render() {
7      ...
8      <li key={item._id} className="menu"><Link to=''{item.name}</Link></li>
9      ...
10     return (
11       ...
12       <li className="menu"><Link to="/">Home</Link></li>
13       ...
14     );
15   }
16   ...
17 }
18 ...

```

- Update 'client-customer/src/components/InformComponent.js' file

```

1  ...
2  import { Link } from 'react-router-dom';
3
4  class Inform extends Component {
5    render() {
6      return (
7        ...
8        <div className="float-left">
9          <Link to=''>Login</Link> | <Link to=''>Sign-up</Link> | <Link to=''>Active</Link>
10         </div>
11         <div className="float-right">
12           <Link to=''>My cart</Link> have <b>0</b> items
13         </div>
14         ...
15       );
16     }
17   }
18   ...

```

- Update 'client-customer/src/components/HomeComponent.js' file

```

1  ...
2  import { Link } from 'react-router-dom';
3
4  class Home extends Component {
5    ...
6    render() {
7      const newprods = this.state.newprods.map((item) => {
8        return (
9          ...
10         <figure>
11           <Link to=''><img ... /></Link>
12           ...
13         </figure>
14         ...
15       );
16     });
17     const hotprods = this.state.hotprods.map((item) => {
18       return (
19         ...
20         <figure>
21           <Link to=''><img ... /></Link>
22           ...
23         </figure>
24         ...
25       );
26     });
27     ...

```

```

28     }
29     ...
30 }
31 ...

```

- Browser: <http://localhost:3001/>

1.1.2 Customer - listproduct

1.1.2.1 Server

- Update 'server/models/ProductDAO.js' file

```

1 ...
2 const ProductDAO = {
3     ...,
4     async selectByCatID(_cid) {
5         const query = { 'category._id': _cid };
6         const products = await Models.Product.find(query).exec();
7         return products;
8     }
9 };
10 ...

```

- Update 'server/api/customer.js' file

```

1 ...
2 // product
3 router.get('/products/category/:cid', async function (req, res) {
4     const _cid = req.params.cid;
5     const products = await ProductDAO.selectByCatID(_cid);
6     res.json(products);
7 });
8 ...

```

- Postman: (GET) <http://localhost:3000/api/customer/products/category/<id>>

1.1.2.2 Client-customer

- Create 'client-customer/src/utlis/withRouter.js' file

```

1 // using withRouter in class-component
2 import { useParams, useNavigate } from "react-router-dom";
3
4 function withRouter(Component) {
5     return (props) => (
6         <Component {...props} params={useParams()} navigate={useNavigate()} />
7     );
8 }
9 export default withRouter;

```

- Create 'client-customer/src/components/ProductComponent.js' file

```

1 import axios from 'axios';
2 import React, { Component } from 'react';
3 import { Link } from 'react-router-dom';
4 import withRouter from '../utlis/withRouter';
5
6 class Product extends Component {
7     constructor(props) {
8         super(props);
9         this.state = {
10             products: []
11         };
12     }
13     render() {
14         const prods = this.state.products.map((item) => {
15             return (

```

```

16     <div key={item._id} className="inline">
17         <figure>
18             <Link to=' '><img src={"data:image/jpg;base64," + item.image} width="300px" height="
19                 300px" alt="" /></Link>
20             <figcaption className="text-center">{item.name}<br />Price: {item.price}</
21                 figcaption>
22         </figure>
23     </div>
24 );
25 });
26 return (
27     <div className="text-center">
28         <h2 className="text-center">LIST PRODUCTS</h2>
29         {prods}
30     </div>
31 );
32 }
33 componentDidMount() { // first: /product/...
34     const params = this.props.params;
35     if (params.cid) {
36         this.apiGetProductsByCatID(params.cid);
37     }
38 }
39 componentDidUpdate(prevProps) { // changed: /product/...
40     const params = this.props.params;
41     if (params.cid && params.cid !== prevProps.params.cid) {
42         this.apiGetProductsByCatID(params.cid);
43     }
44 }
45 // apis
46 apiGetProductsByCatID(cid) {
47     axios.get('/api/customer/products/category/' + cid).then((res) => {
48         const result = res.data;
49         this.setState({ products: result });
50     });
51 }
52 export default withRouter(Product);

```

- Update 'client-customer/src/components/MenuComponent.js' file

```

1 ...
2 class Menu extends Component {
3     ...
4     render() {
5         ...
6         <li key={item._id} className="menu"><Link to={'/product/category/' + item._id}>{item.name
7             }</Link></li>
8         ...
9     }
10    ...
11 }

```

- Update 'client-customer/src/components/MainComponent.js' file

```

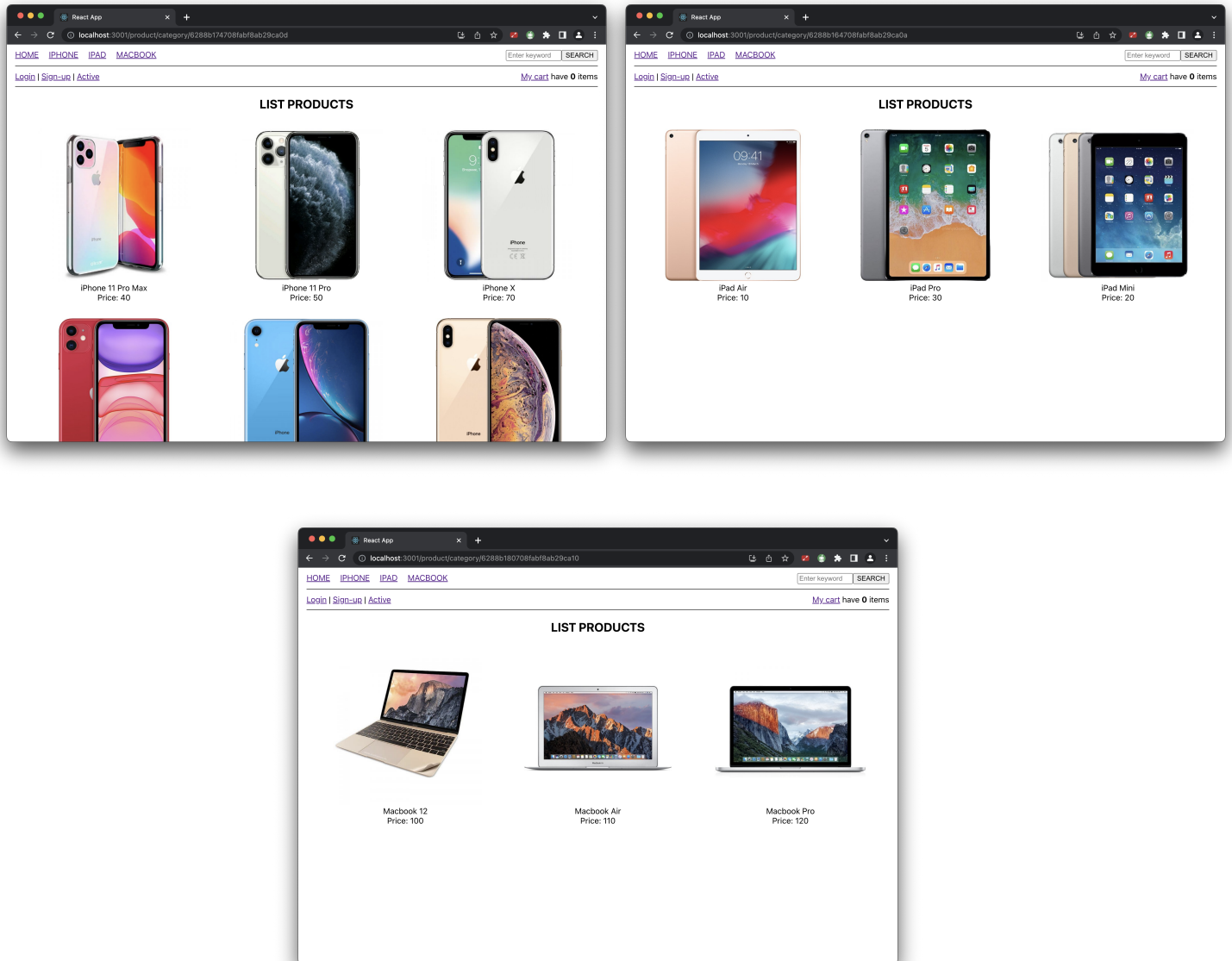
1 ...
2 import Product from './ProductComponent';
3
4 class Main extends Component {
5     render() {
6         return (
7             ...
8             <Routes>
9                 ...
10                <Route path='/product/category/:cid' element={<Product />} />
11            </Routes>
12            ...
13        );
14    }

```

```
15 }  
16 ...
```

- Browser: `http://localhost:3001/product/category/<cid>`

1.1.2.3 Screenshots



1.1.3 Customer - search

1.1.3.1 Server

- Update 'server/models/ProductDAO.js' file

```
1 ...  
2 const ProductDAO = {  
3   ...,  
4   async selectByKeyword(keyword) {  
5     const query = { name: { $regex: new RegExp(keyword, "i") } };  
6     const products = await Models.Product.find(query).exec();  
7     return products;  
8   }  
9 };  
10 ...
```

- Update 'server/api/customer.js' file

```

1  ...
2  // product
3  router.get('/products/search/:keyword', async function (req, res) {
4    const keyword = req.params.keyword;
5    const products = await ProductDAO.selectByKeyword(keyword);
6    res.json(products);
7  });
8  ...

```

- Postman: (GET) <http://localhost:3000/api/customer/products/search/<keyword>>

1.1.3.2 Client-customer

- Update 'client-customer/src/components/MenuComponent.js' file

```

1  ...
2  import withRouter from '../utils/withRouter';
3
4  class Menu extends Component {
5    constructor(props) {
6      ...
7      this.state = {
8        ...,
9        txtKeyword: ''
10     };
11   }
12   render() {
13     ...
14     return (
15       ...
16       <form className="search">
17         <input type="search" placeholder="Enter keyword" className="keyword" value={this.state.
18           txtKeyword} onChange={(e) => { this.setState({ txtKeyword: e.target.value }) }} />
19         <input type="submit" value="SEARCH" onClick={(e) => this.btnSearchClick(e)} />
20       </form>
21       ...
22     );
23   }
24   // event-handlers
25   btnSearchClick(e) {
26     e.preventDefault();
27     this.props.navigate('/product/search/' + this.state.txtKeyword);
28   }
29   ...
30 }
export default withRouter(Menu);

```

- Update 'client-customer/src/components/MainComponent.js' file

```

1  ...
2  class Main extends Component {
3    render() {
4      return (
5        ...
6        <Routes>
7          ...
8          <Route path="/product/search/:keyword" element={<Product />} />
9        </Routes>
10       ...
11     );
12   }
13 }
14 ...

```

- Update 'client-customer/src/components/ProductComponent.js' file

```

1  ...
2  class Product extends Component {

```

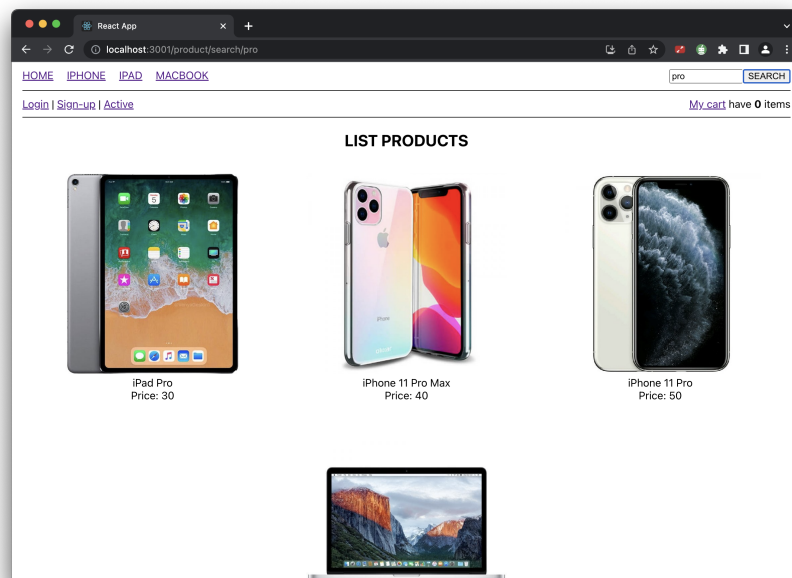
```

3    ...
4    componentDidMount() { // first: /product/...
5        ...
6        if (params.cid) {
7            ...
8        } else if (params.keyword) {
9            this.apiGetProductsByKeyword(params.keyword);
10        }
11    }
12    componentDidUpdate(prevProps) { // changed: /product/...
13        ...
14        if (params.cid && params.cid !== prevProps.params.cid) {
15            ...
16        } else if (params.keyword && params.keyword !== prevProps.params.keyword) {
17            this.apiGetProductsByKeyword(params.keyword);
18        }
19    }
20    // apis
21    apiGetProductsByKeyword(keyword) {
22        axios.get('/api/customer/products/search/' + keyword).then((res) => {
23            const result = res.data;
24            this.setState({ products: result });
25        });
26    }
27    ...
28 }
29 ...

```

- Browser: <http://localhost:3001/> ⇒ enter KEYWORD and press SEARCH button

1.1.3.3 Screenshots



1.1.4 Customer - details

1.1.4.1 Server

- Update 'server/api/customer.js' file

```

1    ...
2    // product
3    router.get('/products/:id', async function (req, res) {
4        const _id = req.params.id;

```

```

5     const product = await ProductDAO.selectByID(_id);
6     res.json(product);
7 });
8 ...

```

- Postman: (GET) <http://localhost:3000/api/customer/products/<id>>

1.1.4.2 Client-customer

- Create 'client-customer/src/components/ProductDetailComponent.js' file

```

1  import axios from 'axios';
2  import React, { Component } from 'react';
3  import withRouter from '../utils/withRouter';
4
5  class ProductDetail extends Component {
6      constructor(props) {
7          super(props);
8          this.state = {
9              product: null
10         };
11     }
12     render() {
13         const prod = this.state.product;
14         if (prod !== null) {
15             return (
16                 <div className="align-center">
17                     <h2 className="text-center">PRODUCT DETAILS</h2>
18                     <figure className="caption-right">
19                         <img src={"data:image/jpg;base64," + prod.image} width="400px" height="400px" alt="
20                             " />
21                         <figcaption>
22                             <form>
23                                 <table>
24                                     <tbody>
25                                         <tr>
26                                             <td align="right">ID:</td>
27                                             <td>{prod._id}</td>
28                                         </tr>
29                                         <tr>
30                                             <td align="right">Name:</td>
31                                             <td>{prod.name}</td>
32                                         </tr>
33                                         <tr>
34                                             <td align="right">Price:</td>
35                                             <td>{prod.price}</td>
36                                         </tr>
37                                         <tr>
38                                             <td align="right">Category:</td>
39                                             <td>{prod.category.name}</td>
40                                         </tr>
41                                         <tr>
42                                             <td align="right">Quantity:</td>
43                                             <td><input type="number" min="1" max="99" /></td>
44                                         </tr>
45                                         <tr>
46                                             <td></td>
47                                             <td><input type="submit" value="ADD TO CART" /></td>
48                                         </tr>
49                                     </tbody>
50                                 </table>
51                             </form>
52                         </figcaption>
53                     </figure>
54                 </div>
55             );
56         }
57         return (<div />);
58     }
59 }

```

```

57     }
58     componentDidMount() {
59         const params = this.props.params;
60         this.apiGetProduct(params.id);
61     }
62     // apis
63     apiGetProduct(id) {
64         axios.get('/api/customer/products/' + id).then((res) => {
65             const result = res.data;
66             this.setState({ product: result });
67         });
68     }
69 }
70 export default withRouter(ProductDetail);

```

- Update 'client-customer/src/components/HomeComponent.js' file

```

1  ...
2  class Home extends Component {
3      ...
4      render() {
5          const newprods = this.state.newprods.map((item) => {
6              return (
7                  ...
8                  <figure>
9                      <Link to={'/product/' + item._id}><img ... /></Link>
10                     ...
11                 </figure>
12                 ...
13             );
14         });
15         const hotprods = this.state.hotprods.map((item) => {
16             return (
17                 ...
18                 <figure>
19                     <Link to={'/product/' + item._id}><img ... /></Link>
20                     ...
21                 </figure>
22                 ...
23             );
24         });
25         ...
26     }
27     ...
28 }
29 ...

```

- Update 'client-customer/src/components/ProductComponent.js' file

```

1  ...
2  class Product extends Component {
3      ...
4      render() {
5          const prods = this.state.products.map((item) => {
6              return (
7                  ...
8                  <figure>
9                      <Link to={'/product/' + item._id}><img ... /></Link>
10                     ...
11                 </figure>
12                 ...
13             );
14         });
15         ...
16     }
17     ...
18 }
19 ...

```

- Update 'client-customer/src/components/MainComponent.js' file

```

1  ...
2  import ProductDetail from './ProductDetailComponent';
3
4  class Main extends Component {
5    render() {
6      return (
7        ...
8        <Routes>
9          ...
10         <Route path='/product/:id' element={<ProductDetail />} />
11       </Routes>
12       ...
13     );
14   }
15 }
16 ...

```

- Browser: <http://localhost:3001/product/<id>>

1.1.4.3 Screenshots

